

Education

- 2018–2022 **PhD in Mathematics**, *Universidad Autónoma de Madrid / ICMAT*, Madrid, *Cum Laude*
PhD thesis on Contact Hamiltonian Systems, under the supervision of Prof. Manuel de León.
- 2017–2018 **Master in Mathematics**, *Universidad Autónoma de Madrid*, Madrid, *Mark: 9.67 / 10*
- 2014–2017 **Degree in Mathematics**, *Universidad de Cantabria*, Santander, *Mark: 8.865 / 10*
Extraordinary Award for the best grades in the class.
- 2012–2016 **Degree in Physics**, *Universidad de Cantabria*, Santander, *Mark: 8.81 / 10*
- 2015–2016 **Exchange program during undergraduate studies**, *Princeton University*, Princeton
I took courses in both Mathematics and Physics, and wrote my Physics undergraduate thesis.

Research experience

- 2023– **Professor (Profesor Contratado Doctor)**, *CUNEF University*, Madrid, Spain
Teaching and research, full time.
- 2022–2023 **Postdoctoral researcher**, *ICMAT–CSIC*, Madrid, Spain
- 2019–2022 **FPI–Severo Ochoa predoctoral contract**, *ICMAT–CSIC*, Madrid, Spain
PhD thesis on Contact Hamiltonian Systems, under the supervision of Prof. Manuel de León.
- 2018–2019 **Predocctoral contract (PhD program)**, *Universidad Autónoma de Madrid / ICMAT*, Madrid, Spain
- 2017–2018 **Research grant (Beca de colaboración, Severo Ochoa / María de Maeztu)**, *ICMAT–CSIC*, Madrid, Spain
Applications of contact geometry to mechanics, under the supervision of Prof. Manuel de León.

Research stays

- 2018 **Department of Mathematics, Universitat Politècnica de Catalunya**, Barcelona, Spain
Two-week research stay (project PGC2018-098265-B-C33): geometry of contact systems with dissipation, unified formalism, inverse problem and optimal control of the Herglotz principle.
- 2015–2016 **Princeton University**, Princeton, USA
Year-long exchange during the undergraduate degree.

Teaching experience

- 2023– **Linear Geometry, Linear Algebra and Optimization**, *CUNEF University*, Madrid
Degree in Mathematical Engineering and double degree in Economics and Mathematical Engineering.
- 2021–2022 **Statistics applied to the environment**, *Universidad Autónoma de Madrid*, Madrid, Spain
Degree in Environmental Sciences and double degree in Environmental Sciences and Land Planning.
- 2019–2020 **Algebra**, *Universidad Autónoma de Madrid*, Madrid, Spain
Degree in Computer Engineering.
- 2019–2020 **Linear Algebra**, *Universidad Autónoma de Madrid*, Madrid, Spain
Degree in Telecommunications Technologies and Services Engineering.
- 2017 **Teaching assistant, Física Cuántica III (Solid State Physics)**, *Universidad de Cantabria*, Santander, Spain
Degree in Physics.
Total university teaching: approx. 789 hours.

Research grants and projects

- 2022 **Juan de la Cierva postdoctoral grant (JDC2022-049361-I)**, *Ministerio de Ciencia*, Spain
Awarded; declined due to incompatibility with the current contract.
- 2023–2025 **Geometry of Dynamical Systems: from Theory to Applications I**, *ICMAT*
Member of the research team (PI: David Martín de Diego).
- 2020–2023 **Geometric Structures in Dynamical Systems, Mechanics and Hydrodynamics**, *ICMAT*
Member of the research team (PI: David Martín de Diego and Daniel Peralta-Salas).

Supervision

- 2026 **Master's thesis: "Geometric Mechanics of Generalized Disformal Transformations"**, *Matteo Dell'Acqua*
Elite MSc Program in Theoretical and Mathematical Physics (LMU & TU München), Munich, Germany.

Accreditations and merits

- 2025 ANECA accreditation as Profesor Contratado Doctor and as Profesor de Universidad Privada (Ciencias Experimentales I).
- 2023 ANECA accreditation as Profesor Ayudante Doctor (Ciencias Experimentales I).
- 2019–2024 Recognition of a six-year research period (*sexenio*) by ANECA and CUNEF.

Languages

Spanish Native

English Advanced (C1)

*103 points on T.O.E.F.L.
Advanced level at the Spanish Escuela Oficial de Idiomas
Trinity level 8
1 year of studies at Princeton University*

■ Outreach

- 2025 Speaker at the European Researchers' Night (Horizon Europe / Marie Skłodowska-Curie Actions), interactive science-communication session "La naturaleza es contraintuitiva. Matemática y física para sorprendernos".
- 2021 Participant in the 1st #HiloTesis science-communication contest (CRUE).

■ Other merits

- 2021–2022 **Organizer of the Joint Junior Colloquium (ICMAT–UAM–UC3M–UCM)**
Biweekly colloquium aimed at PhD students and young researchers.
- 2016 Bronze Medal at the International Mathematics Competition for University Students, Blagoevgrad, Bulgaria.

Bio

Manuel Lainz is currently working as a Professor in CUNEF. With a background in Mathematics (where he obtained an end of degree award for the best grades in his year) and Physics from Universidad de Cantabria and a one-year study period at Princeton University.

He continued his Master's studies in Mathematics at UAM while doing an internship at ICMAT under the supervision of Manuel de León, where he started working on differential geometry and its applications to physics and other sciences. He stayed on ICMAT for his PhD thesis at, again under the guidance of Professor Manuel de León, focused on Contact Hamiltonian systems and their use in modeling irreversible dynamics. He focused on developing tools for understanding and controlling the behavior of thermodynamic and mechanical systems with friction. His contributions to the theory of Contact Hamiltonian systems have led to numerous advancements both theoretically and practically, including the editor-picked publication "Contact Hamiltonian Systems" in the Journal of Mathematical Physics, which has received 164 citations on Google Scholar, and other 4 articles with more than 50 citations. He has also collaborated with prominent authors in the field, both nationally and internationally, on various topics such as plasma physics, control theory, and thermodynamics. After his thesis was defended, he continued to make advancements in several areas of geometric mechanics, including field theories, Hamilton-Jacobi theory, integrable systems, and quantization of nonholonomic systems.

Since 2023, he holds a position at CUNEF. His research has moved to the study of several generalizations of integrability, including the Arnold-Liouville theorem for contact systems and of PDEs through multisymplectic geometry.

He has presented my research findings and delivered lectures and mini-courses at several international conferences.